

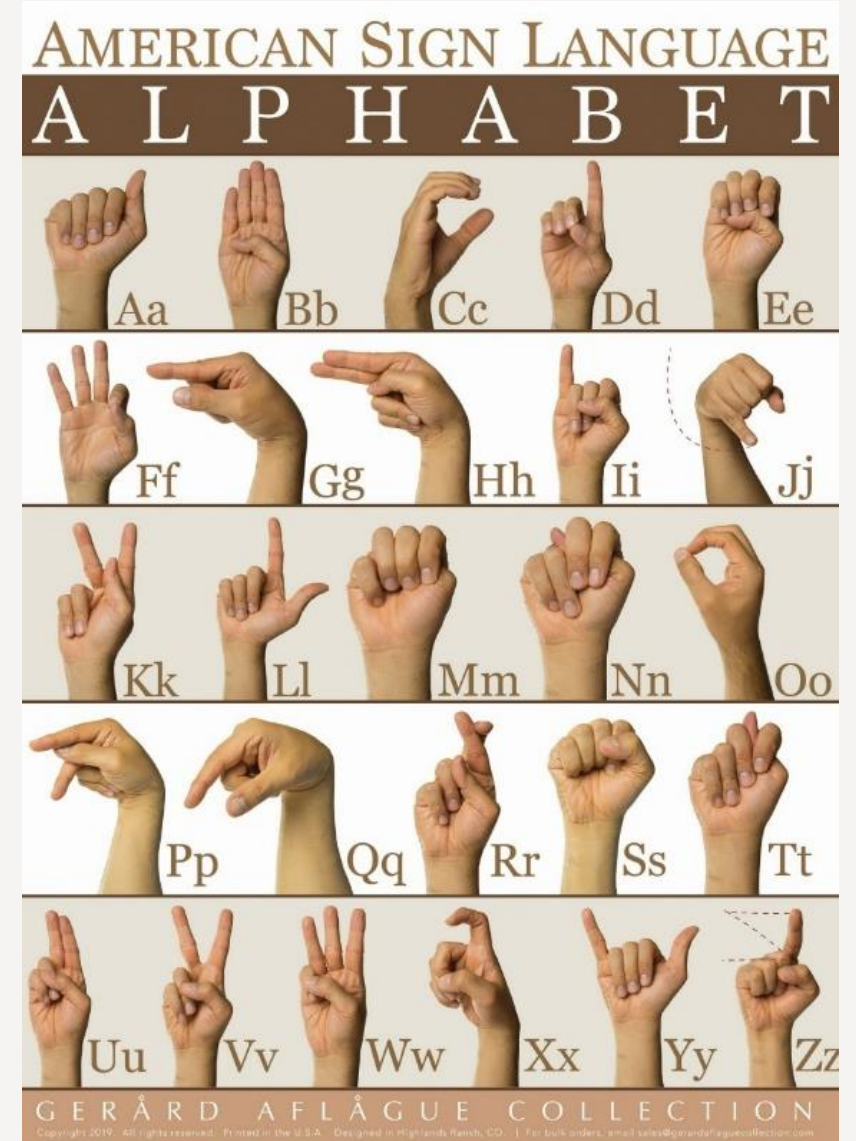
DATS 6303 – Deep Learning

American Sign Language (ASL) Alphabet Classification Using Deep Learning

A comparative study of ResNet-50 and
MobileNetV2

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Source: Atangan, K. (2019, February 17)

Estimated **1 million Deaf and Hard-of-Hearing population** use sign language in the United States

[Rosen et al., 2022]

Sign language use approaches **3% of the U.S. adult population**

[Rosen et al., 2022]

Classify static ASL alphabet handshapes from RGB images using deep learning

Why it matters

- Helps recognize ASL signs automatically.
- Supports better communication accessibility.

Selected problem

- Serves as a first step toward full sign-language recognition systems
- Useful for assistive and educational applications.

Final outcome

- Compare two transfer-learning models and deploy a webcam-based Streamlit demo

Scope: Letters A–Z

Dynamic signing and full sentence recognition are outside this project.

26

letter classes

Two public ASL datasets were combined to create a balanced A–Z dataset.

Kaggle ASL Alphabet

87K
Kaggle
images

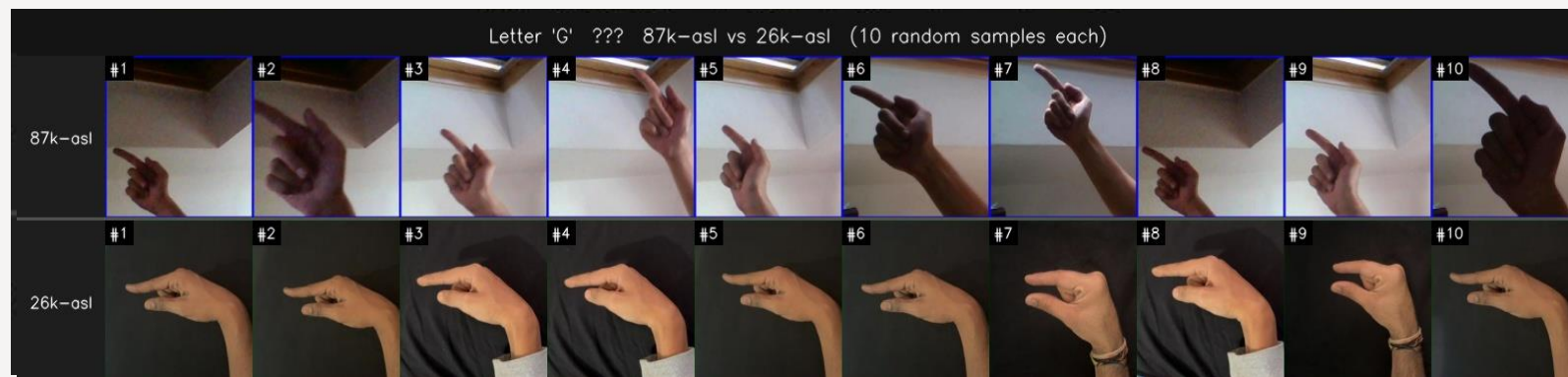
29
raw Kaggle
classes

Mendeley ASL Dataset

26K
Mendeley
images

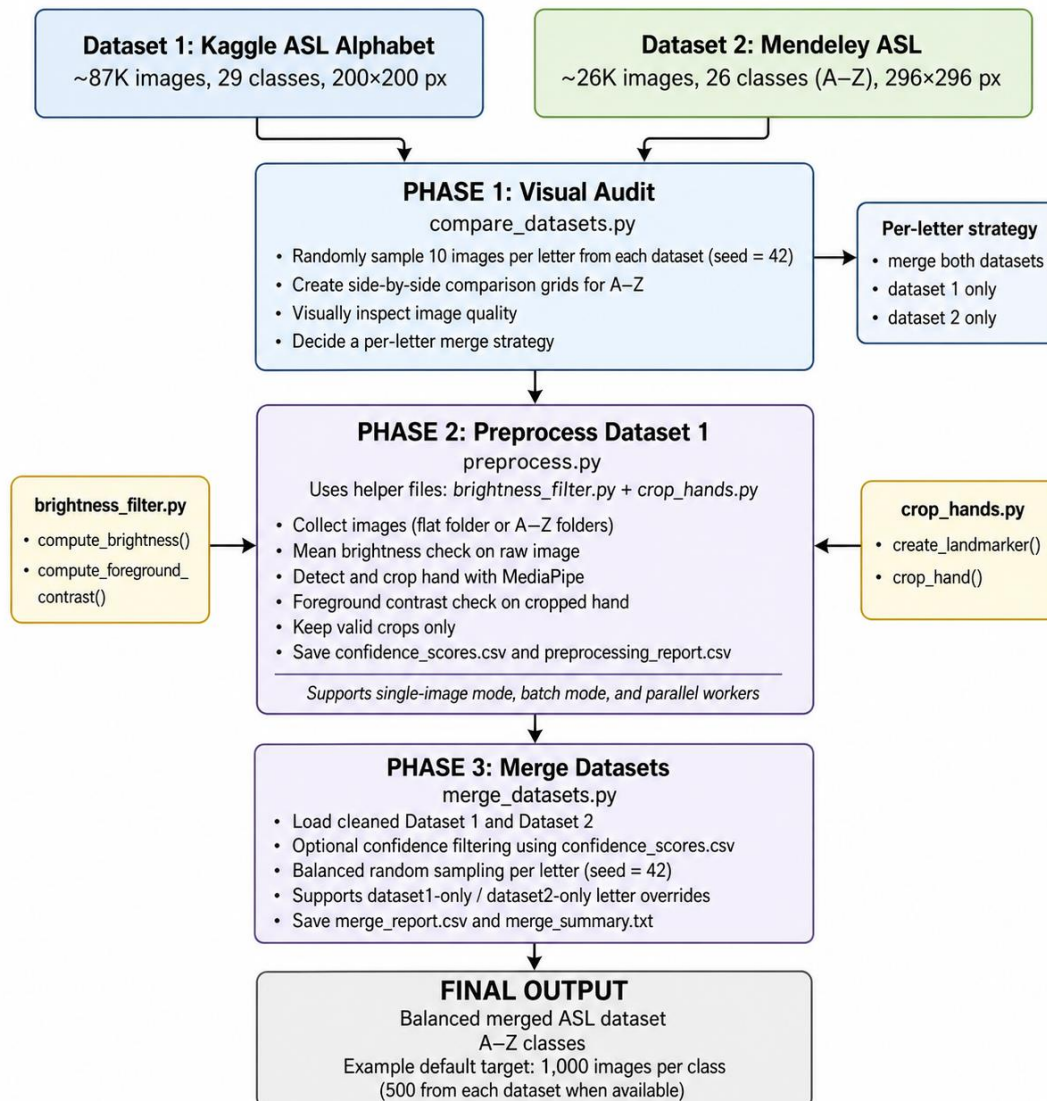
26
A–Z
classes

Pilot finding: Kaggle dataset → alone achieved high training accuracy but poor real-time webcam inference because of wide-angle images, low contrast, label noise, and hidden hand features.



M: hidden features
G: label contamination

Three-phase dataset preprocessing pipeline



Phase 1: Visual audit

Checked compatibility by orientation, hand shape, and lighting

Phase 2: Preprocess 87K

Applied brightness filtering, hand cropping, and contrast checks

Phase 3: Merge

Created a balanced 26K-image dataset with 1,000 images per class

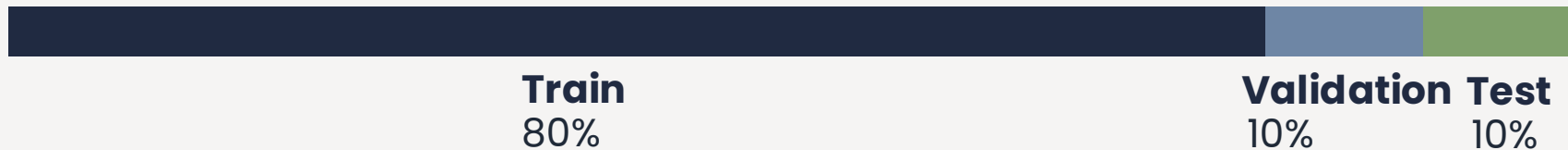
Final split and Model Selection

26,000
total images

26
letter classes

1,000
images per class

The standardized dataset is balanced across 26 A–Z letter classes and split with a fixed random seed.



CHOSEN MODELS

ResNet-50

- Strong baseline model for high image-classification accuracy.
- Good at learning complex hand-shape features.

MobileNetV2

- Lightweight and faster, so it is easier to deploy
- Better suited for a Streamlit demo and real-time use.

Two-phase transfer learning adapts pretrained feature extractors to the ASL alphabet task.



4. Build Model

Pretrained ResNet50 (ImageNet)
+ custom classification head



5. Phase 1 Training

Freeze ResNet50 base,
train classifier head, Adam lr = 0.001



6. Phase 2 Fine-Tuning

Unfreeze top ResNet50 layers (from layer 140),
Adam lr = 0.00001

Implemented the network in TensorFlow/Keras

Modified ResNet50 for ASL classification

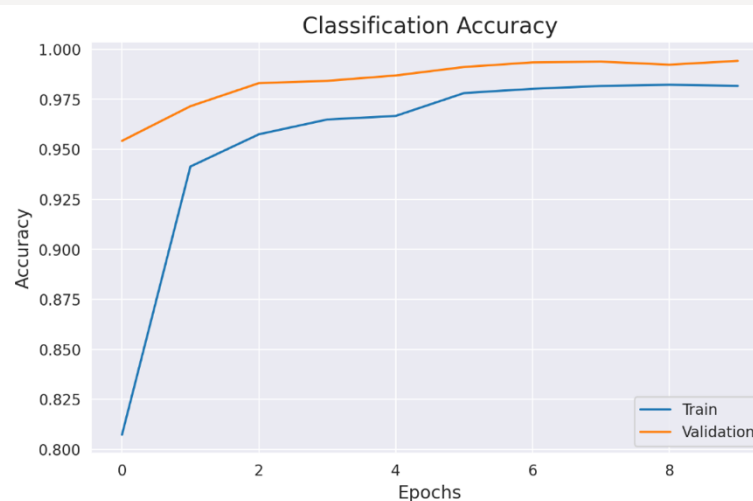
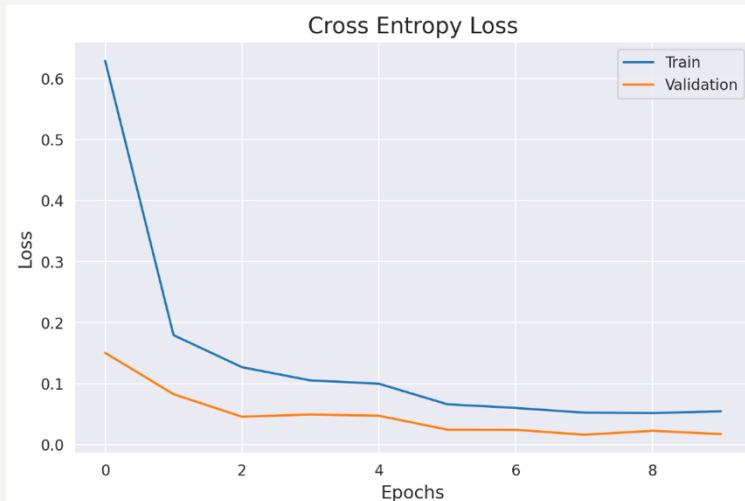
Phase 1 training:

To allow the model to learn how to map ResNet50 image features to the ASL letters

Phase 2 fine-tuning:

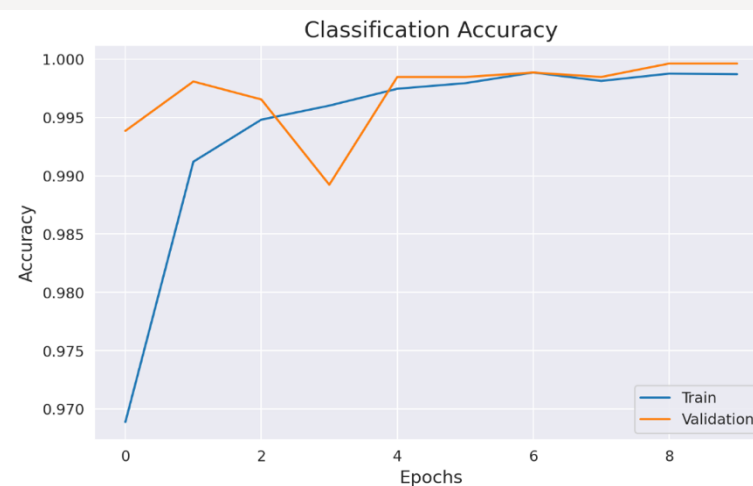
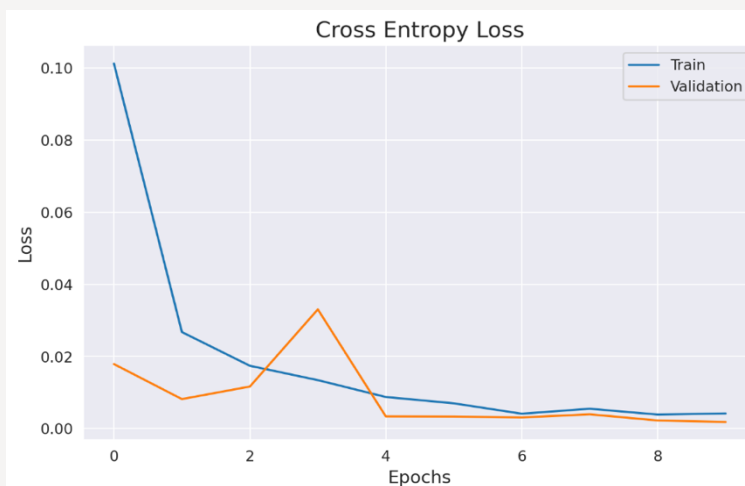
To allow the later layers to adjust to the ASL dataset

Phase 1 — frozen base



Training phase	Final val. accuracy	Final val. loss
Phase 1	99.42%	0.0173
Phase 2	99.96%	0.0018

Phase 2 — fine-tuning



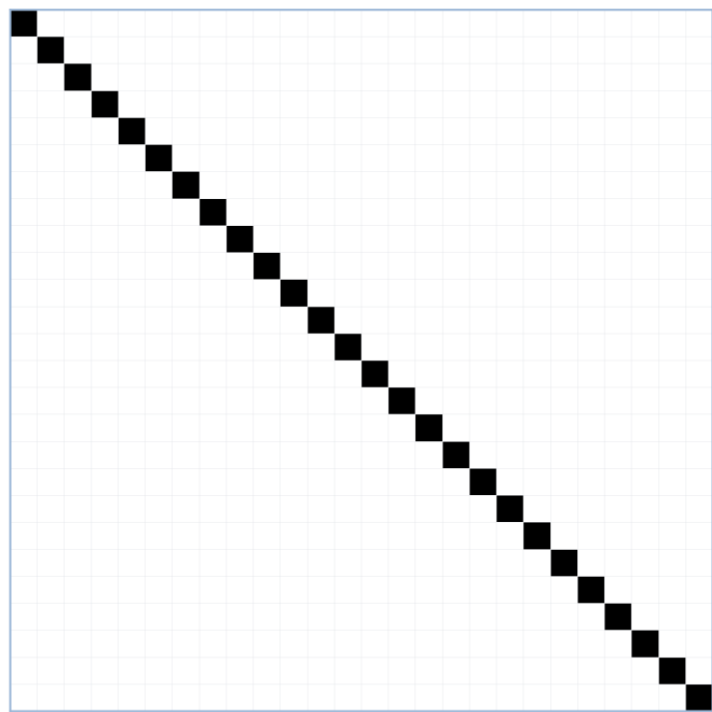
Overall accuracy

99.99%

Test loss: 0.0003

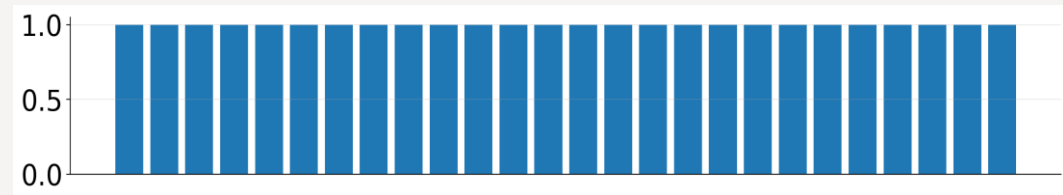
No visible class confusion: all 26 letters have perfect per-class scores.

Confusion matrix pattern



Only diagonal cells are filled: each letter was predicted as itself.

Per-class F1 pattern



Check	Result
Test images	2,600
Support per class	100
Confused pairs	None
Per-class F1	1.00 for A–Z

MobilenetV2 Training algorithm

Two-phase transfer learning adapts pretrained feature extractors to the ASL alphabet task.

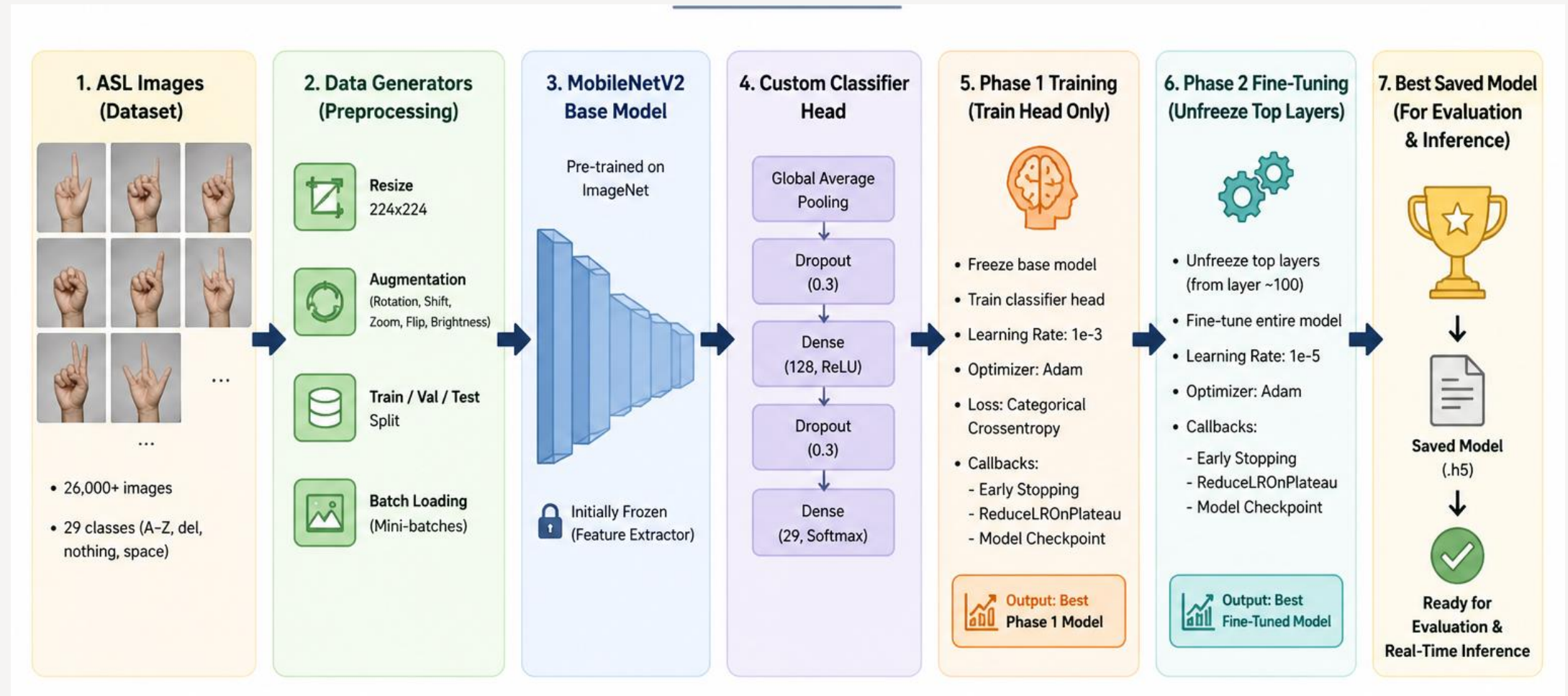
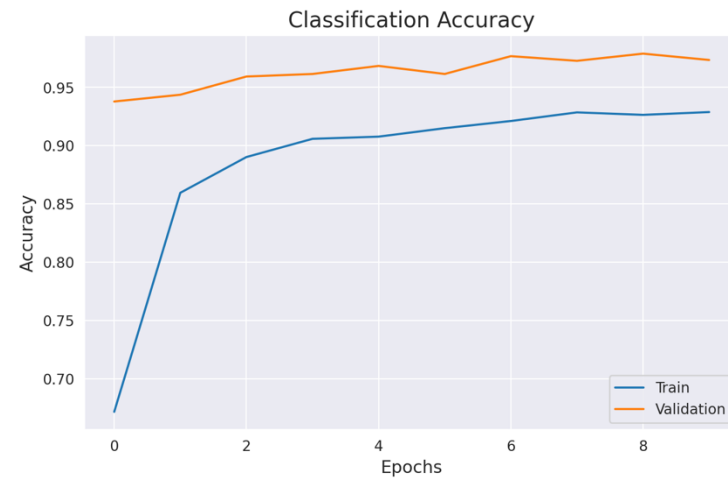
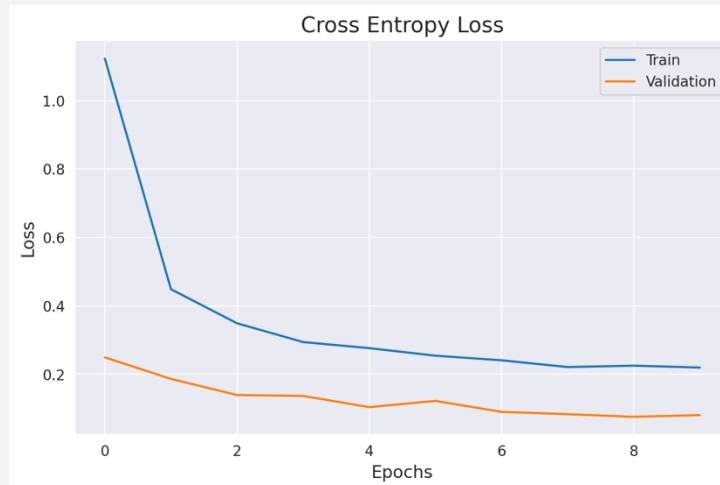


Diagram was created using Generative AI

Phase 1 Curve

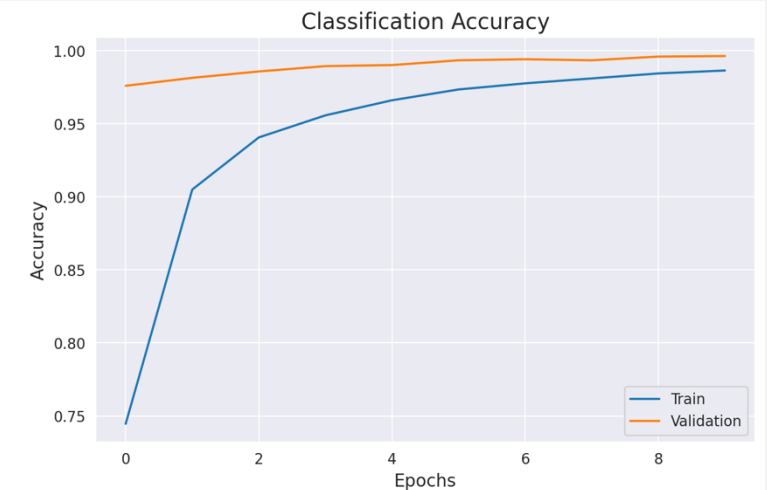


Phase 1 accuracy:
~67% → ~93%

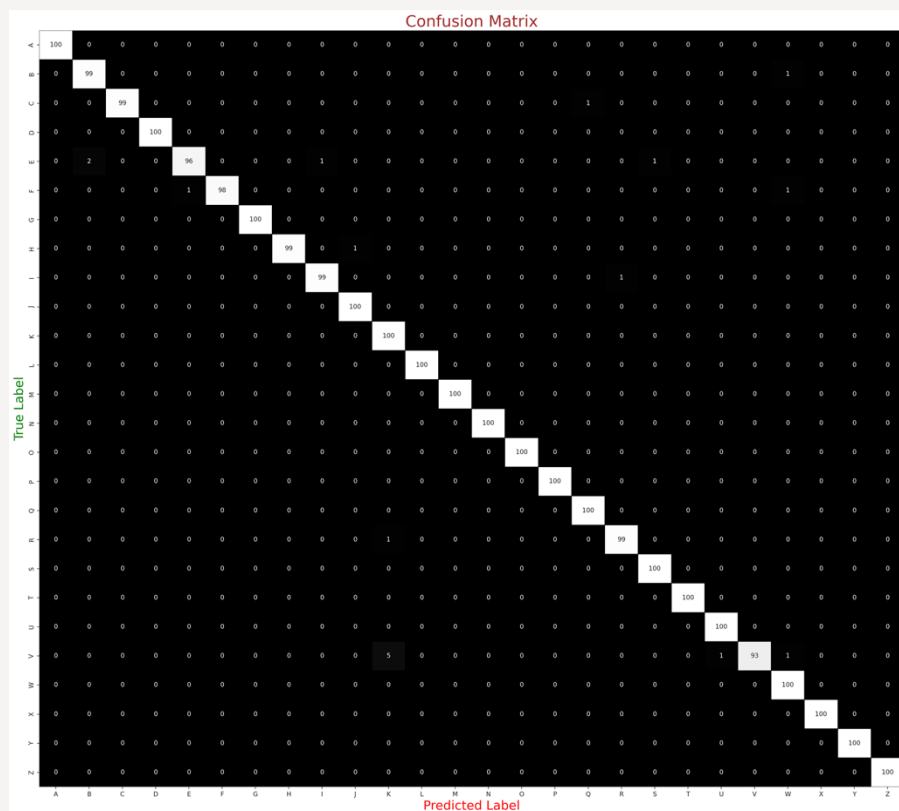
Phase 2 Curve

Phase 2 accuracy:
~74% → ~98.6%

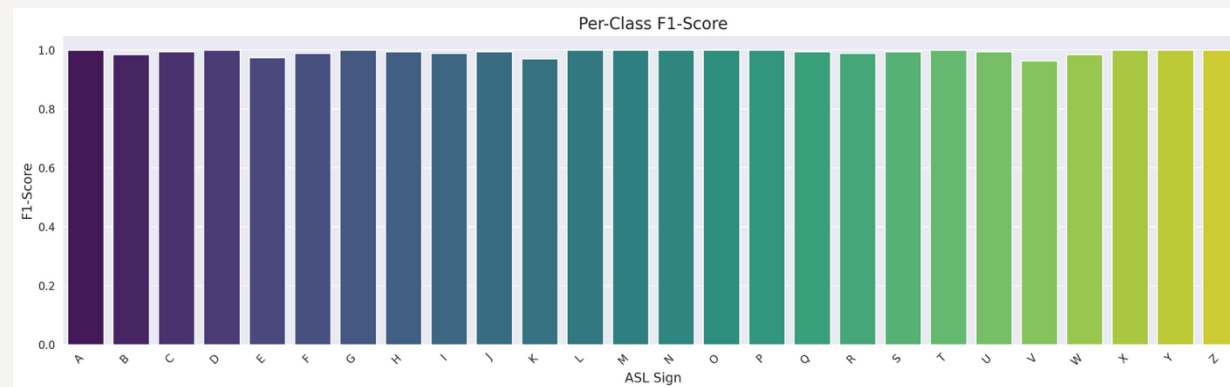
Validation accuracy
reached ~99.6%



Confusion matrix pattern



Per Class F1-score



Metrics Card

TEST ACCURACY



99.67%

Top-1 Accuracy

MACRO F1-SCORE



99.69%

Macro F1-Score

TOP-5 ACCURACY



100.00%

Top-5 Accuracy

TEST LOSS



0.0102

Cross Entropy Loss

PER-CLASS PERFORMANCE



~99%+

Most classes near perfect F1-score

ResNet-50

- **Peak Accuracy: 99.71%**
- **10x Larger**
- **Heavier and Slower**

MobileNetV2

- **Peak Accuracy: 99.67%**
- **Lightweight & Efficient**
- **Faster Interface**

Higher capacity



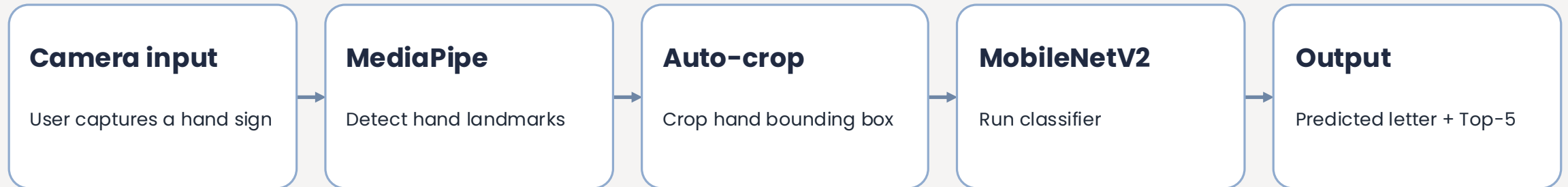
ResNet-50

More deployable



MobileNetV2

MobileNetV2 is selected for demo deployment because it is lightweight and efficient



Why crop?

Full webcam frames contain background, lighting variation, and hand-size mismatch. Cropping reduces domain shift.

User-facing output

The app displays the predicted class, confidence, cropped hand image, and Top-5 probabilities.

- Both models achieved **near-perfect performance (~99.6% accuracy)**
- **ResNet50 slightly higher accuracy (+0.04%)** but with **~10× more parameters**
- Lower performance on **M, N, T, Y**
 - **due to visually similar hand shapes & subtle finger differences**
- **Data augmentation + dataset merging**
 - **improved robustness and real-world performance**

Limitations & Future Work

- Static images only (no motion/sequence)
- Struggles with similar gestures
- Future: **video-based models, larger datasets, Vision Transformers**

- Akash (2018), Kaggle ASL Alphabet Dataset; Benabbas (2022), Mendeley ASL Dataset.
- He et al. (2016), Deep Residual Learning for Image Recognition — ResNet-50 backbone.
- Sandler et al. (2018), MobileNetV2: Inverted residuals and linear bottlenecks.
- Lugaresi et al. (2019), MediaPipe framework for hand landmark detection.
- Rosen, R. S., Henner, J., & Villwock, A. (2022). How many people use sign language? A national health survey-based estimate. Journal of Deaf Studies and Deaf Education, 28(1), 1–11.
<https://doi.org/10.1093/jdsde/enac032>
- TensorFlow / Keras, OpenCV, scikit-learn, Split-Folders, and Streamlit used in implementation.

Project repository: <https://github.com/ashleygya03/FinalProject-Group6>

THANKS!

Questions?

